

Analysis of Quantization Methods in Transformer-Based Neural Networks

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Research questions and contribution

Research questions

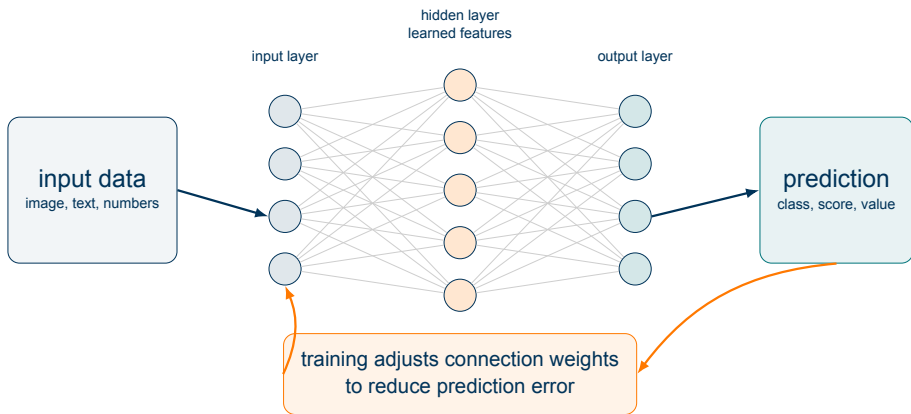
1. How do bit width and method affect accuracy and estimated storage?
2. How do the three saved checkpoints respond at W4 and W8?
3. Which kernels, tensors or modules are most sensitive?
4. Does a W8 sensitivity ranking transfer to W4?

What was implemented

- Two custom post-training weight quantizers
- 42 full-model sweep configurations
- W4/W8 isolated sensitivity analysis
- Saved predictions and permutation tests

The study compares complete checkpoint–quantizer configurations and does not isolate architecture as the cause.

What is a neural network?

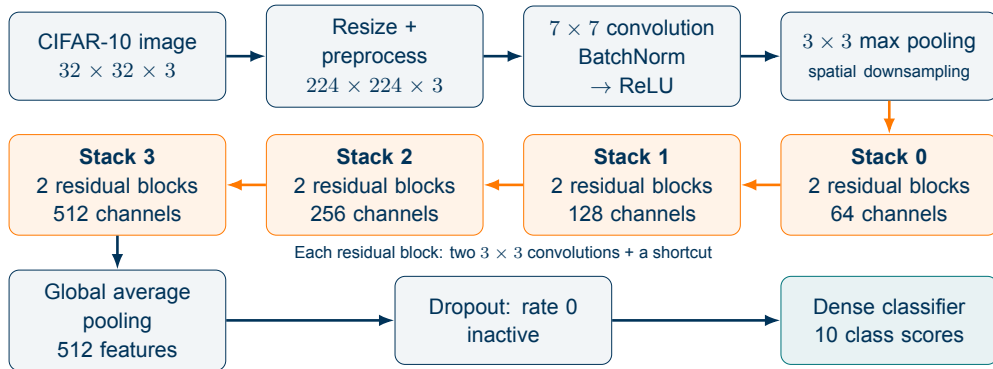


A neural network learns weighted connections that transform input data into a useful prediction.

What is a convolutional neural network?

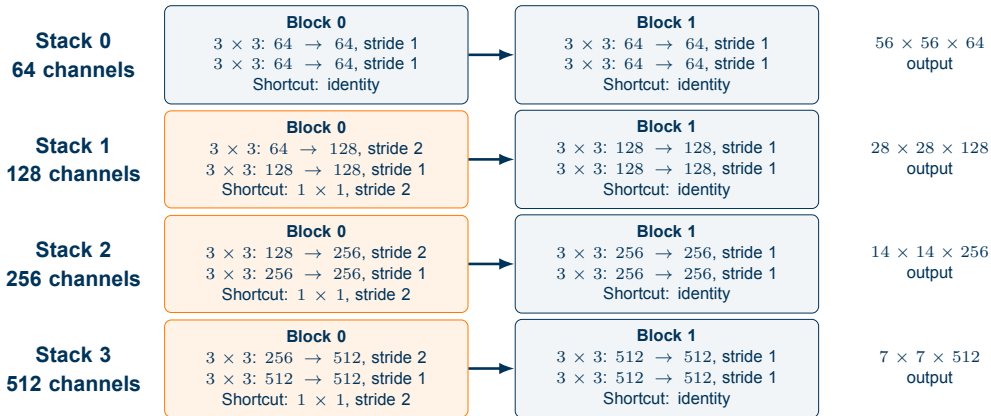
A CNN learns filters that slide across an image, detecting local patterns and combining them into higher-level features.

ResNet-18 used in this study



ResNet-18: expanded stacks 0–3

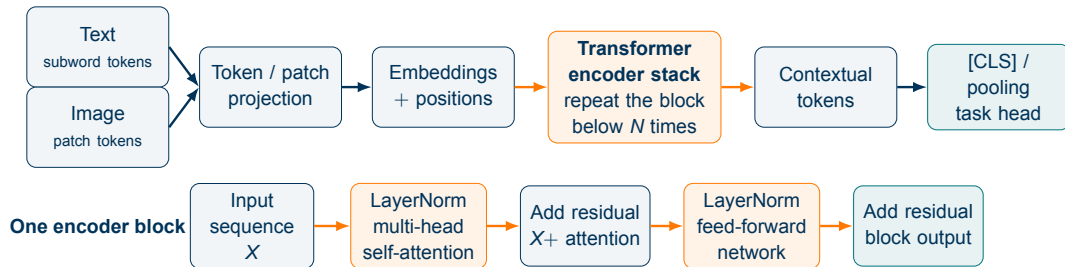
Each stack contains two residual blocks; each convolution is followed by BatchNorm.



Block: Conv → BN → ReLU → Conv → BN → add shortcut → ReLU. Projection shortcuts also use BN.

What is a Transformer?

A Transformer represents an input as a sequence of tokens, uses self-attention to exchange information between them, and transforms each token with a feed-forward network.

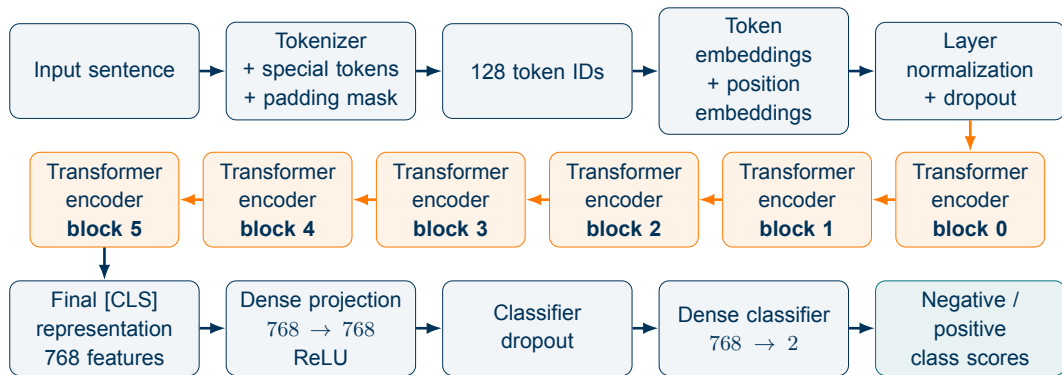


Text uses subword tokens; vision uses image-patch tokens. Both use the same encoder computation.

Vaswani et al., 2017

What is DistilBERT?

DistilBERT is a smaller version of BERT learned through knowledge distillation. Its six encoder blocks build contextual text representations for tasks such as sentiment classification.

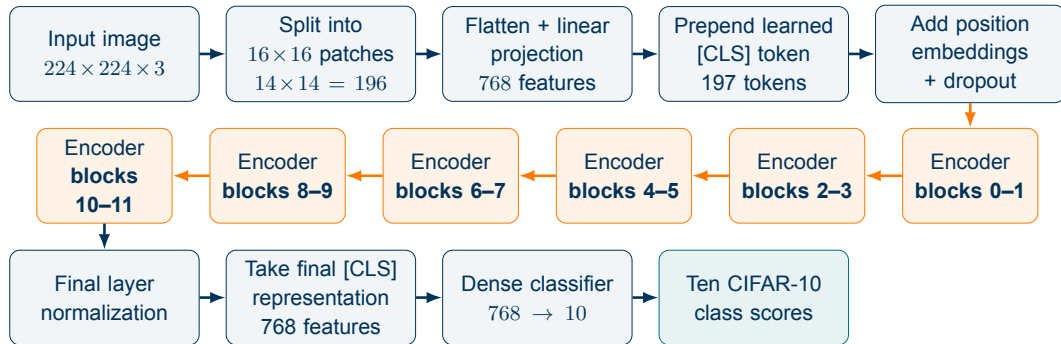


SST-2: 872 evaluation examples; 88.42% FP32 accuracy.

Sanh et al., 2019

What is ViT-B/16?

ViT-B/16 is a Vision Transformer that converts an image into 16×16 -pixel patch tokens and processes them with 12 Transformer encoder blocks for image classification.



CIFAR-10: 10,000 test images; 95.73% FP32 accuracy.

Dosovitskiy et al., 2021

What is quantization?

Quantization represents high-precision neural-network values with a smaller set of low-bit discrete values.

What changes?

FP32 weights are approximated using b -bit values.

Why use it?

Fewer bits reduce model storage and memory traffic.

What is the cost?

Rounding introduces error that may reduce accuracy.

Scope of this study

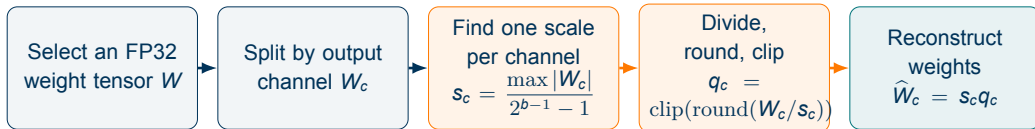
Both custom methods are post-training and weight-only, with no quantization-aware training.
Estimated packed parameter storage; accuracy evaluated using FP32-reconstructed weights and FP32 activations.

The objective is to reduce the bits per weight while retaining as much FP32 accuracy as possible.

Uniform PTQ: method and implementation

Uniform post-training quantization maps trained FP32 weights to an evenly spaced low-bit grid after training, without gradient updates or retraining.

Jacob et al., 2018



Configuration used here

$b = 2, 3, \dots, 8$; signed symmetric grid;
 $2^b - 1$ levels; per-output-channel scales.

How it was evaluated

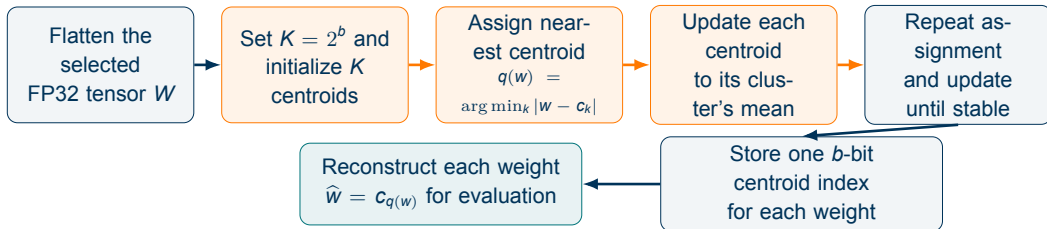
Weight-only simulated PTQ: reconstructed weights and FP32 activations were used for accuracy evaluation.

Each weight is replaced by its nearest available grid value; lower bit widths create a coarser approximation.

Codebook quantization: how it works

Codebook quantization learns $K = 2^b$ representative weight values, called centroids, and replaces every FP32 weight with the index of its nearest centroid.

Gray, 1984; Lloyd, 1982



Cluster means follow the tensor's weight distribution, so the learned centroid levels are generally unevenly spaced.

The codebook contains the centroid values; the quantized tensor contains their indices.

Experimental methodology: tensor selection



same selection for PTQ and codebook

ResNet-18: 21

All floating tensors with rank ≥ 2
Convolution and classifier kernels

DistilBERT: 40

Paths ending in `/kernel` or `/embeddings`
Token/position embeddings; Q, K, V and attention-output kernels; both feed-forward kernels; pooled and classifier kernels

ViT-B/16: 76

Paths ending in `/kernel` or `/embeddings`, plus `class_token`
Patch/position embeddings; six kernels in each of 12 blocks; classifier kernel

For sensitivity analysis, the 76 ViT tensors are grouped into 14 logical modules; seven predefined modules are tested at W4.

No bias tensors were selected; rank-one biases, normalization scale/offset vectors and rank-two attention biases remained FP32.

Evaluated checkpoints and datasets

Model	Dataset	Examples	Parameters	FP32 size	Accuracy
ResNet-18	CIFAR-10 test	10,000	11.19 M	42.69 MiB	90.47%
DistilBERT	SST-2 dev	872	66.96 M	255.41 MiB	88.42%
ViT-B/16	CIFAR-10 test	10,000	85.81 M	327.33 MiB	95.73%

ResNet-18

Adapted end-to-end

DistilBERT

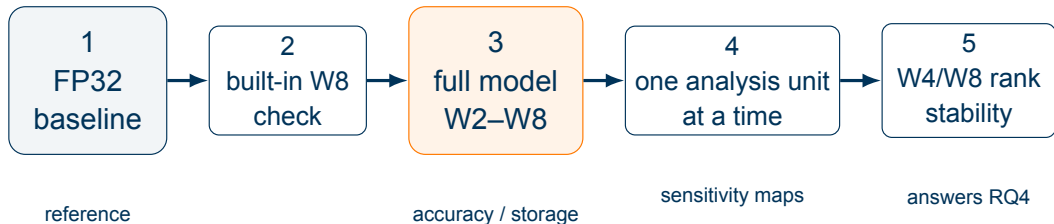
Fine-tuned for SST-2

ViT-B/16

Backbone frozen; head trained

Every quantized result is compared with the FP32 accuracy of the same saved checkpoint.

Experimental methodology: experiment map



Full-model tensors

ResNet: 21 DistilBERT: 40 ViT: 76

Isolated sensitivity

W8: exhaustive W4: 21 kernels / 15 tensors / 7 modules

The sweep measures system behavior; isolated replacement locates which weights drive it.

Baseline checks

Model	Built-in path	FP32	Built-in W8	Compression
ResNet-18	Full INT8, W8A8	90.47%	88.53%	4.00×
DistilBERT	TFLite dynamic-range quantization	88.42%	88.07%	3.99×
ViT-B/16	TFLite dynamic-range quantization	95.73%	95.78%	3.96×

Custom PTQ: full W2–W8 sweep

A: accuracy (%)

R: compression ratio ($\times$)

S: parameter-memory reduction (%)

Bits	ResNet-18			DistilBERT			ViT-B/16		
	A	R	S	A	R	S	A	R	S
2	10.00	15.60	93.59	51.03	15.67	93.62	13.00	15.50	93.55
3	11.23	10.49	90.47	82.91	10.52	90.49	32.22	10.45	90.43
4	11.56	7.91	87.35	88.99	7.92	87.37	95.23	7.88	87.31
5	49.39	6.34	84.23	88.19	6.35	84.25	95.83	6.32	84.19
6	71.87	5.29	81.11	88.07	5.30	81.13	95.70	5.28	81.06
7	88.05	4.54	77.99	88.30	4.55	78.01	95.75	4.53	77.94
8	89.88	3.98	74.87	88.42	3.98	74.88	95.79	3.97	74.82

Estimated packed parameter storage; accuracy uses FP32-reconstructed weights.

Scalar codebook: full W2–W8 sweep

A: accuracy (%)

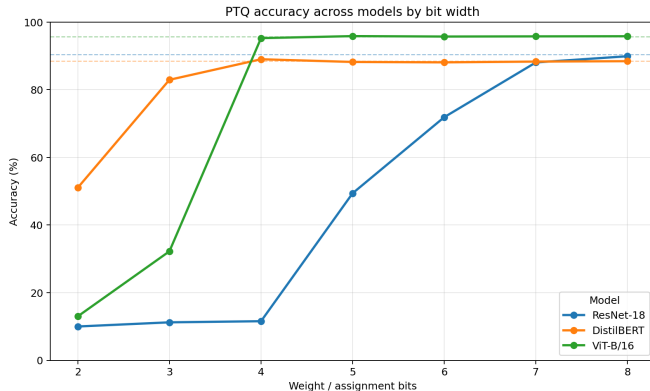
R: compression ratio ($\times$)

S: parameter-memory reduction (%)

Bits	ResNet-18			DistilBERT			ViT-B/16		
	A	R	S	A	R	S	A	R	S
2	9.50	15.60	93.59	78.33	15.78	93.66	20.84	15.66	93.62
3	10.00	10.49	90.47	86.24	10.57	90.54	89.77	10.52	90.50
4	8.43	7.90	87.35	88.30	7.95	87.42	95.20	7.92	87.37
5	15.41	6.34	84.22	88.19	6.37	84.29	95.64	6.35	84.25
6	48.71	5.29	81.10	88.07	5.31	81.17	95.76	5.30	81.13
7	88.97	4.54	77.97	88.42	4.55	78.04	95.68	4.55	78.00
8	90.25	3.97	74.82	88.07	3.99	74.92	95.73	3.98	74.87

Estimated packed parameter storage; accuracy uses FP32-reconstructed weights.

Results: custom PTQ accuracy sweep



At W4

ResNet-18
11.56%

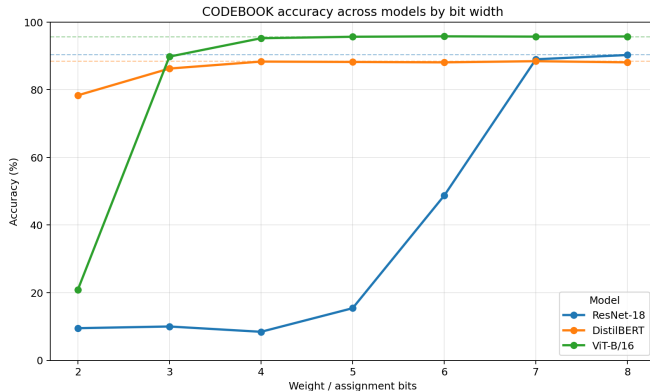
DistilBERT
88.99%

ViT-B/16
95.23%

Observed transition

ResNet recovers near W7; both transformer checkpoints recover at W4.

Comparison: custom codebook accuracy sweep



Low-bit retention

DistilBERT W2
78.33%

ViT-B/16 W3
89.77%

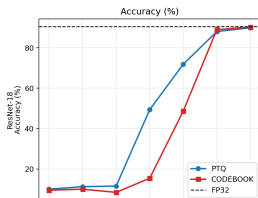
ResNet-18 W4
8.43%

Method contrast

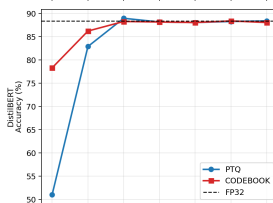
Largest observed gap: ViT W3, codebook exceeds PTQ by 57.55 pp. The configurations also differ in granularity and level count.

Accuracy and storage trade-off

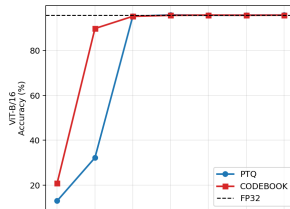
ResNet-18



DistilBERT



ViT-B/16



Storage is mainly bit-width-driven

Bits	Compression	Reduction
2	15.5–15.8×	93.6%
4	7.9×	87.3–87.4%
8	4.0×	74.8–74.9%

Estimated packed parameter storage.

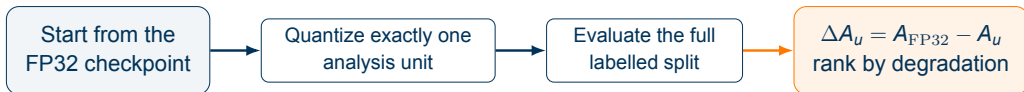
One storage result, two views

$$S = 100 (1 - 1/R)$$

Memory reduction S and compression ratio R are mathematically linked.

Accuracy uses FP32-reconstructed weights.

Layer-by-layer comparison



Controlled comparison

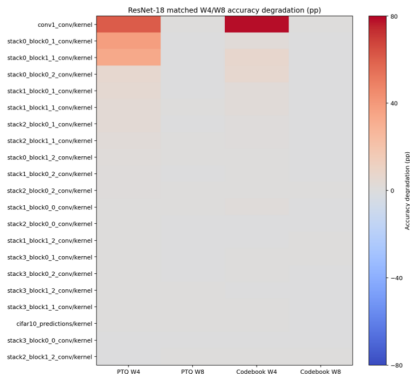
Every run restores the original checkpoint and leaves all other weights in FP32. Positive ΔA_u is an accuracy loss; a negative value is an observed gain.

Model	Unit	W8	W4
ResNet-18	kernel	21/21	21/21
DistilBERT	tensor	40/40	15/40
ViT-B/16	module	14/14	7/14

Heatmaps: red = loss, blue = observed gain.

This experiment locates vulnerable units; it does not measure interactions between several quantized units.

ResNet-18: layer sensitivity



Colour range: ± 80 pp

Three most sensitive at W4

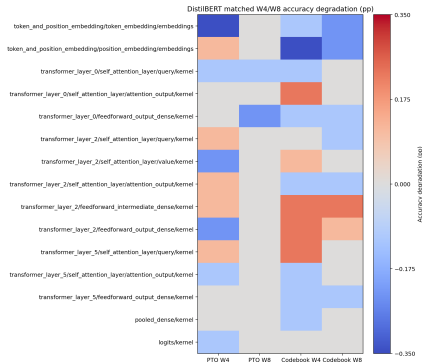
1. **First convolution**
78.62 pp codebook; 60.65 pp PTQ
2. **Stack 0, block 0, conv 1**
37.72 pp PTQ
3. **Stack 0, block 1, conv 1**
33.19 pp PTQ

Explanation

The common ± 80 -pp scale exposes the extreme early-kernel losses; the smaller effects appear pale. At W8, every isolated loss is at most 0.53 pp.

The W4 failure is strongly localized to early kernels, especially the first convolution.

DistilBERT: tensor sensitivity



Colour range: ± 0.35 pp

Three most sensitive at W4

Codebook losses; tied at 0.229 pp and ordered by prediction disagreement:

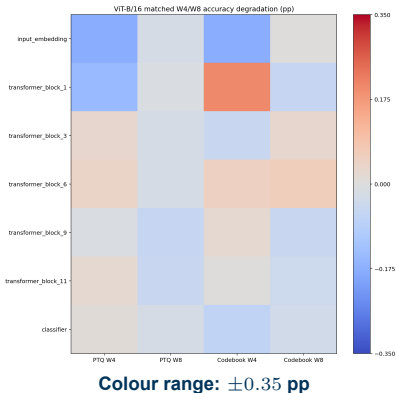
1. Layer 2 intermediate FFN
2. Layer 0 attention output
3. Layer 2 output FFN

Explanation

Effects are small and distributed across attention and feed-forward tensors. W4 covers 15 of 40 tensors.

No sampled tensor shows a ResNet-like collapse; the small effects must not be read as statistical equivalence.

ViT-B/16: module sensitivity



Three most sensitive at W4

1. **Transformer block 1**
0.20 pp codebook
2. **Transformer block 6**
0.05 pp codebook; 0.04 pp PTQ
3. **Transformer block 3**
0.03 pp PTQ

Explanation

The largest sampled effect occurs early, but all seven W4 module effects stay within ± 0.20 pp.

Overall: W8 is robust across these checkpoints; lower-bit accuracy is model- and method-dependent. ResNet's W4 failure concentrates in early kernels, while the sampled transformer units remain close to FP32.

Experiment 5: does W8 sensitivity predict W4? (RQ4)

Model	Method	Units	ρ	p_{perm}
ResNet-18	PTQ	21	-0.094	0.68527
ResNet-18	Codebook	21	-0.292	0.19867
DistilBERT	PTQ	15	0.143	0.61139
DistilBERT	Codebook	15	0.693	0.00551
ViT-B/16	PTQ	7	-0.321	0.49762
ViT-B/16	Codebook	7	-0.357	0.44444

One positive result

DistilBERT codebook

$$\rho = 0.693$$

$$p_{\text{perm}} = 0.00551$$

ρ : Spearman correlation between the W8 and W4 sensitivity rankings.

p_{perm} : permutation-test probability of observing an association this strong if the rankings are unrelated.

Interpretation

A significant positive rank association was observed for the tested DistilBERT codebook subset. The other five tests provide insufficient evidence of transfer.

RQ4: General W8-to-W4 ranking transfer was not established.

Comparison: observed pattern and interpretation boundary

	ResNet-18	DistilBERT	ViT-B/16
W8	near baseline	near baseline	near baseline
W4	severe collapse	within 0.6 pp	within 0.6 pp
Codebook edge	no at W4	large at W2–W3	very large at W3
Sensitive region	early kernels	scattered tensors	selected modules

W8 headline

≤ 0.59 **pp loss**
approximately $4\times$ estimated compression

Transformer W4 headline

within 0.6 pp
approximately $7.9\times$ estimated compression

These are checkpoint- and pipeline-specific observations; training and quantizer differences prevent an architecture-only conclusion.

The strongest defensible claim concerns these three evaluated configurations, not transformers versus CNNs in general.

Conclusions: answers to the research questions

RQ1: How do bit width and method affect accuracy and storage?

Compression followed bit width (about $4\times$ at W8 and $7.9\times$ at W4), while accuracy depended on the checkpoint and method; codebook retained more low-bit accuracy in selected DistilBERT and ViT cases.

RQ2: How do the checkpoints respond at W4 and W8?

At W8, all three checkpoints remained within 0.59 pp of FP32; at W4, ResNet-18 collapsed to 11.56% PTQ and 8.43% codebook, while DistilBERT and ViT stayed within 0.6 pp.

RQ3: Which units are most sensitive?

ResNet-18's W4 damage concentrated in early kernels, especially `conv1`; the sampled DistilBERT tensors and ViT modules produced only small isolated accuracy changes.

RQ4: Does the W8 sensitivity ranking transfer to W4?

General transfer was not established; only DistilBERT codebook was significant ($\rho = 0.693$, permutation $p = 0.00551$, Bonferroni-adjusted $p = 0.03306$).

Selected references

1. K. He et al., “Deep Residual Learning for Image Recognition,” CVPR, 2016.
2. A. Vaswani et al., “Attention Is All You Need,” NeurIPS, 2017.
3. V. Sanh et al., “DistilBERT,” arXiv:1910.01108, 2019.
4. A. Dosovitskiy et al., “An Image Is Worth 16x16 Words,” ICLR, 2021.
5. B. Jacob et al., “Quantization and Training of Neural Networks,” CVPR, 2018.
6. R. Krishnamoorthi, “Quantizing Deep Convolutional Networks,” arXiv:1806.08342, 2018.
7. R. M. Gray, “Vector Quantization,” IEEE ASSP Magazine, 1984.
8. S. P. Lloyd, “Least Squares Quantization in PCM,” IEEE TIT, 1982.

Statistical and transformer-PTQ references are listed in the accompanying report.

Questions?

Thank you

TU Ilmenau campus



Backup: BatchNorm hypothesis for the ResNet W4 collapse



Recorded policy

- BatchNorm scale, offset, moving mean and variance remain FP32.
- Inference uses *training=False*.
- No folding, recalibration, bias correction, equalization or adaptive rounding.

What is established

- Quantizing *conv1* alone causes the largest W4 loss.
- BatchNorm mismatch is a plausible mechanism, but it was not isolated or tested.
- The study makes no claim that recalibration would recover accuracy.

The location of the failure is measured; its causal mechanism remains unresolved.